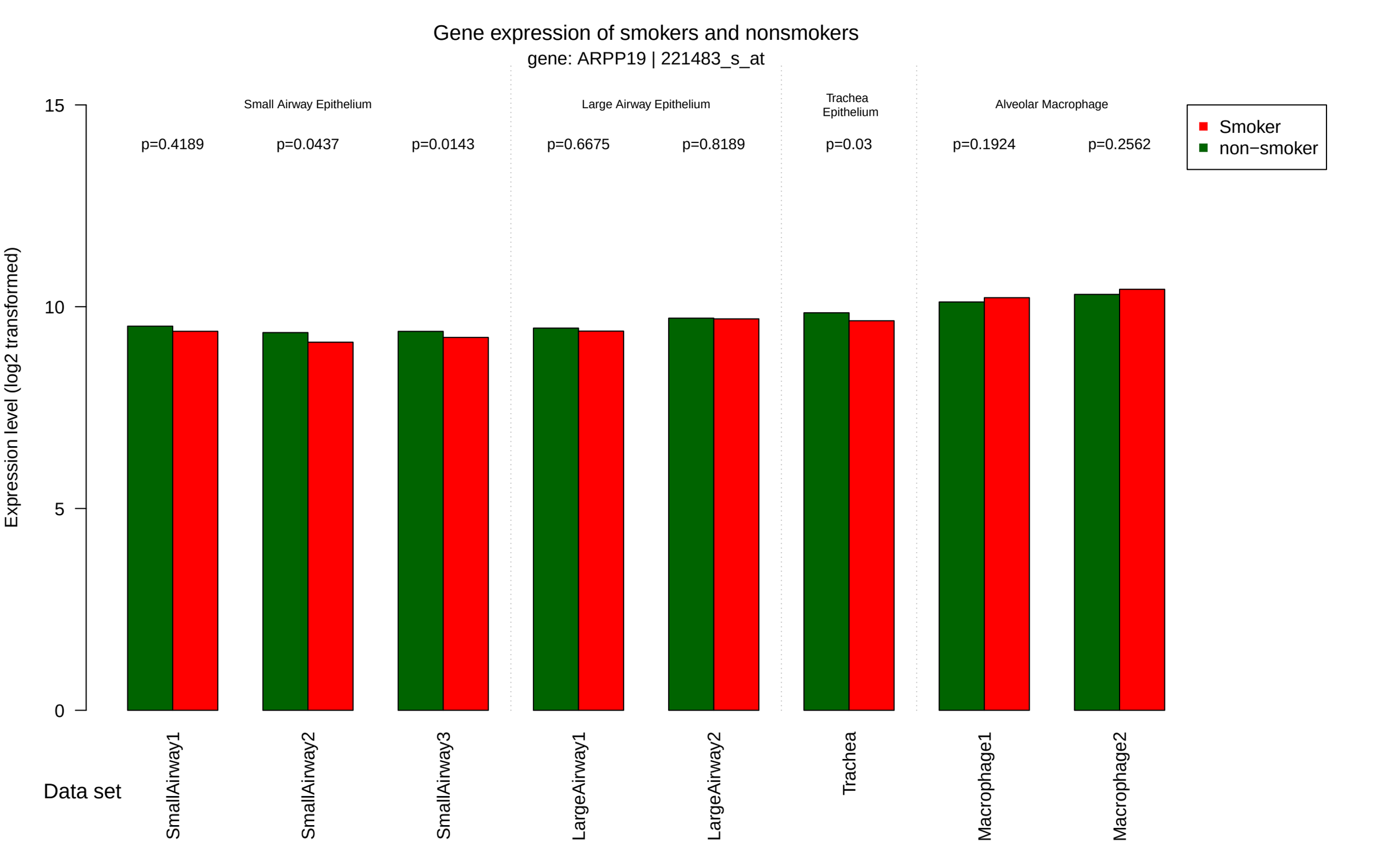
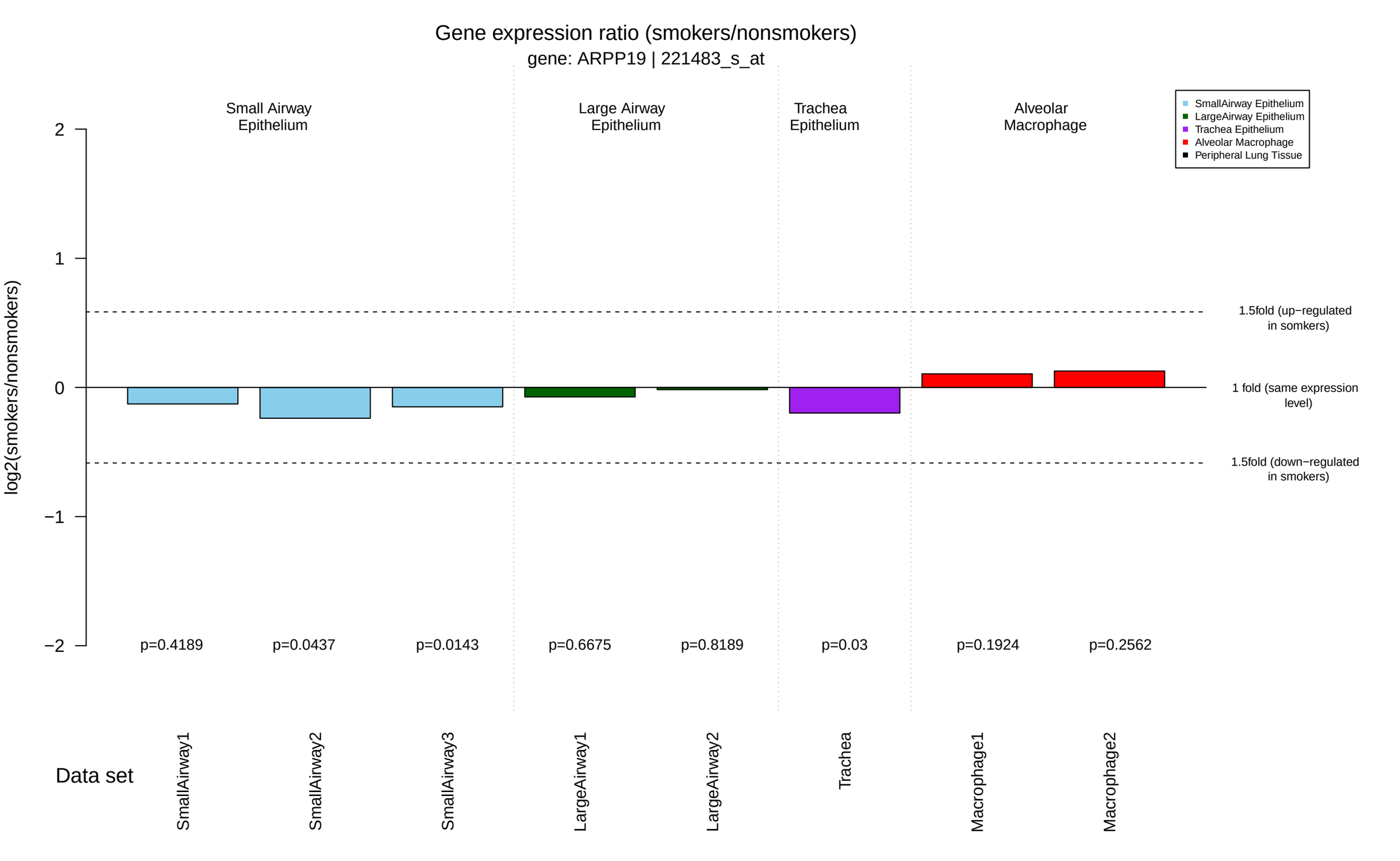
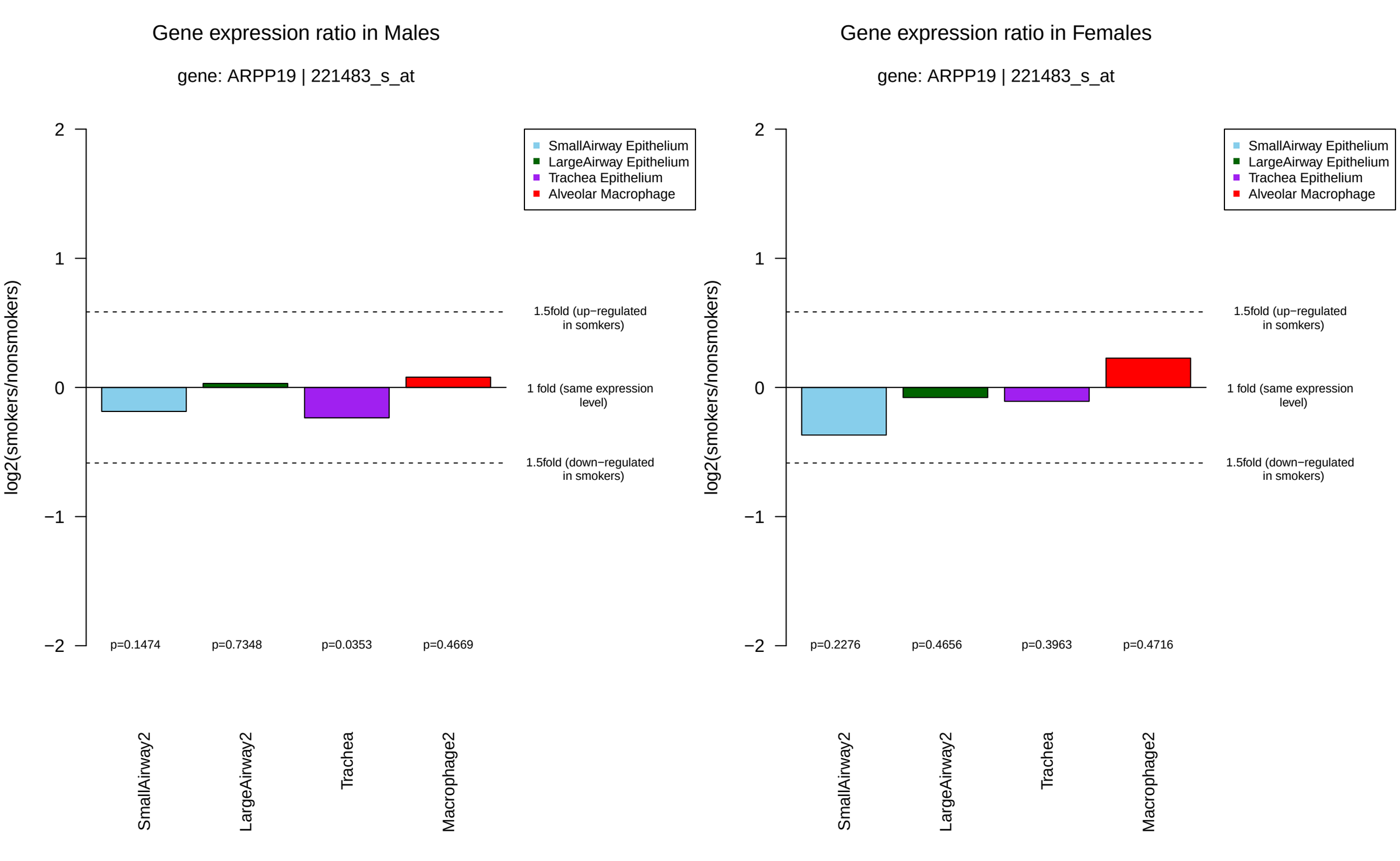
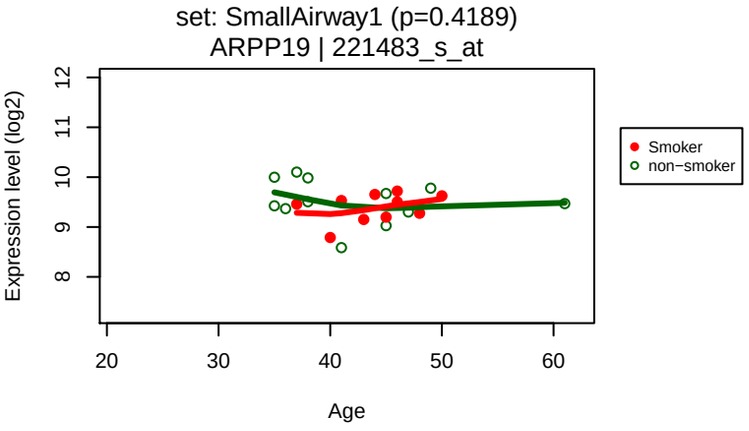
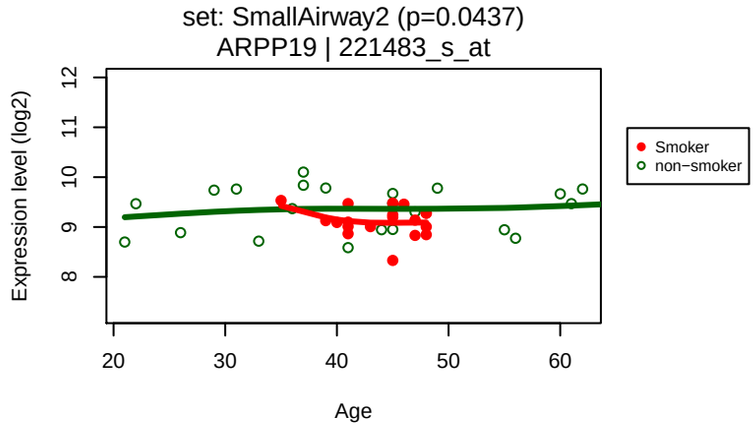
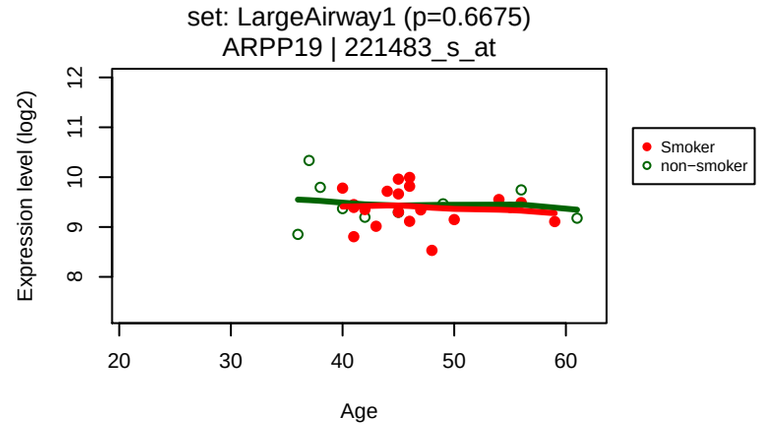

Gene expression of smokers and nonsmokers
gene: ARPP19 | 221483_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4189
p=0.0437
p=0.0143
p=0.6675
p=0.8189
p=0.03
p=0.1924
p=0.2562
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

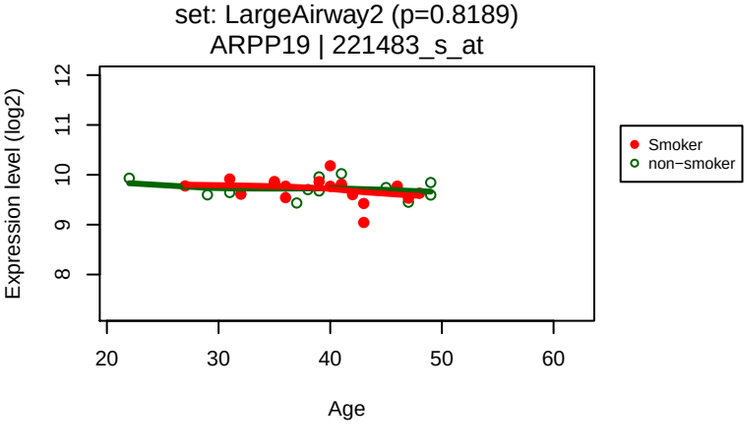

Gene expression ratio (smokers/nonsmokers)
gene: ARPP19 | 221483_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4189
p=0.0437
p=0.0143
p=0.6675
p=0.8189
p=0.03
p=0.1924
p=0.2562
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

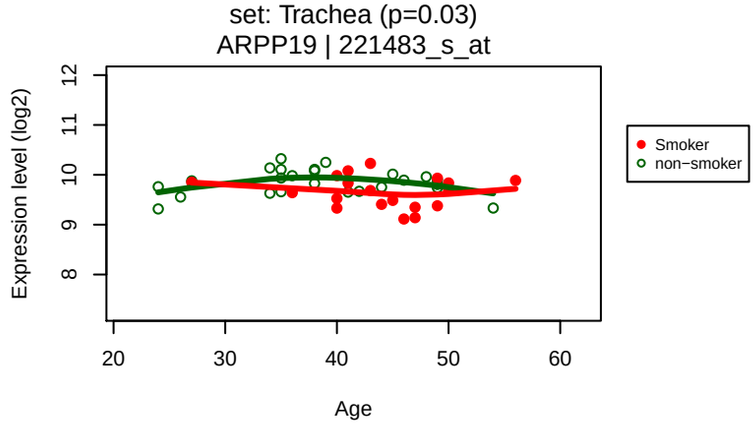

Gene expression ratio in Males
gene: ARPP19 | 221483_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1474
p=0.7348
p=0.0353
p=0.4669
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ARPP19 | 221483_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2276
p=0.4656
p=0.3963
p=0.4716
SmallAirway2
LargeAirway2
Trachea
Macrophage2

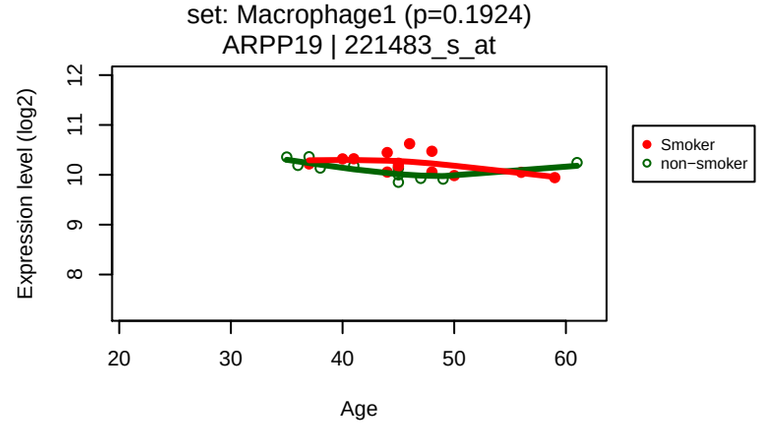

set: SmallAirway1 (p=0.4189)
ARPP19 | 221483_s_at
Expression level (log2)
Age
Smoker
non−smoker

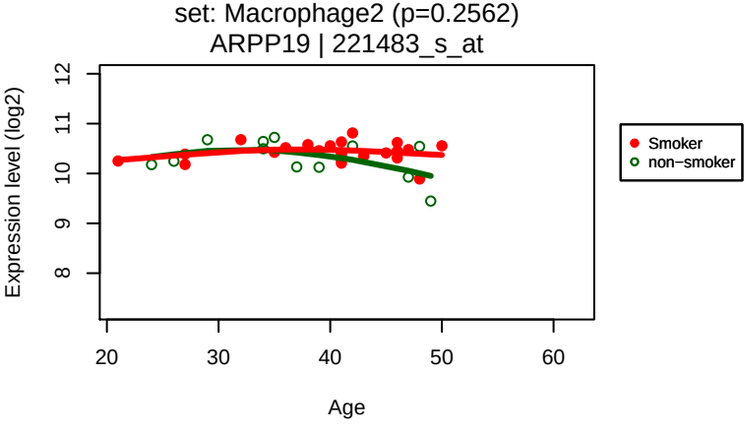

set: SmallAirway2 (p=0.0437)
ARPP19 | 221483_s_at
Expression level (log2)
Age
Smoker
non−smoker

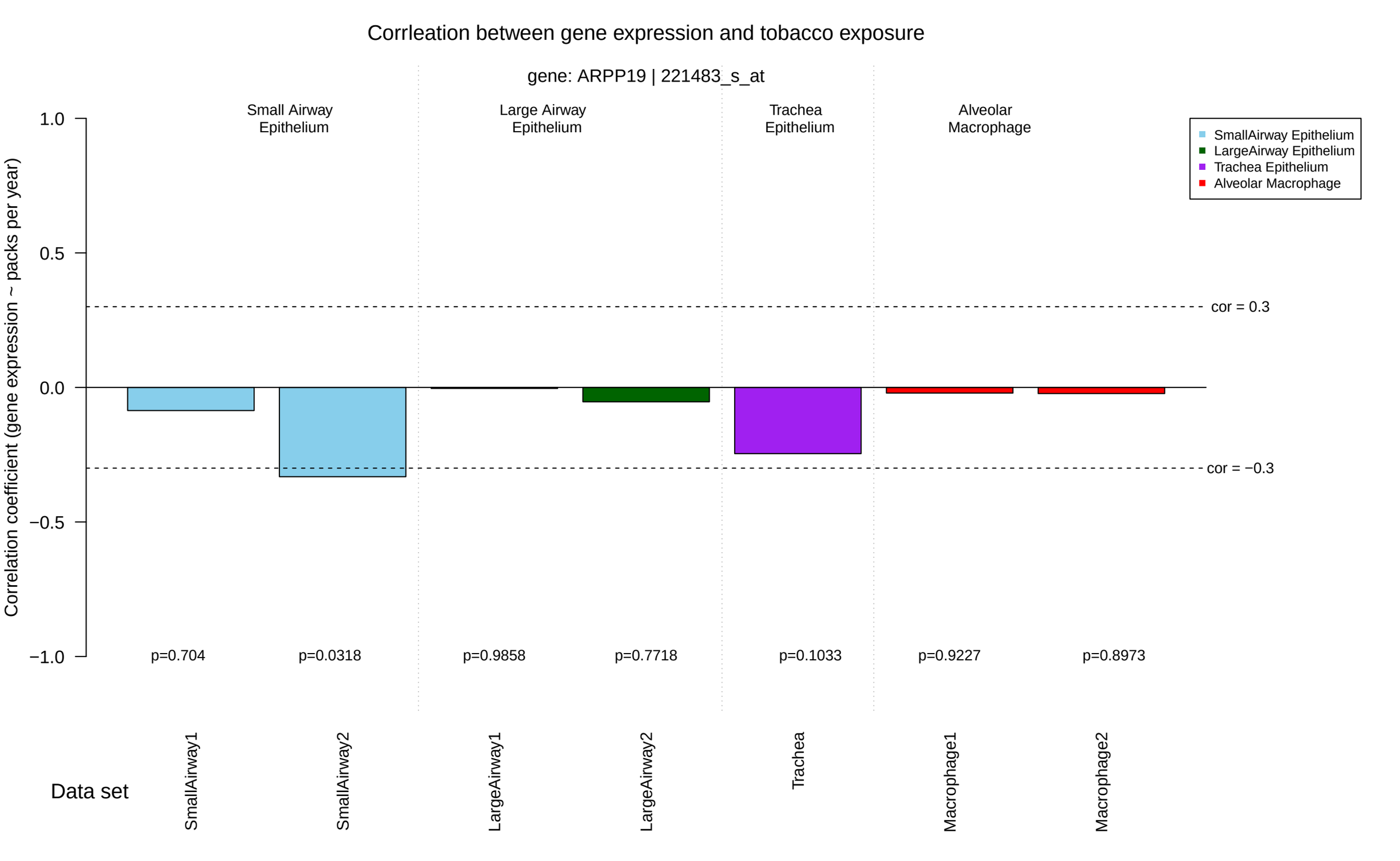

set: LargeAirway1 (p=0.6675)
ARPP19 | 221483_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8189)
ARPP19 | 221483_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.03)
ARPP19 | 221483_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1924)
ARPP19 | 221483_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2562)
ARPP19 | 221483_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ARPP19 | 221483_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.704
p=0.0318
p=0.9858
p=0.7718
p=0.1033
p=0.9227
p=0.8973
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2